



TYPICAL PROPERTIES of Radel® (PPSU)				
ASTM or UL test	Property	Radel® unfilled (A-300)	Radel® 30% glass filled (AG-330)	Radel® R
PHYSICAL				
D792	Density (lb/in ³) (g/cm ³)	0.050 1.37	0.057 1.58	0.047 1.30
D570	Water Absorption, 24 hrs (%)	0.54	0.39	0.4
MECHANICAL				
D638	Tensile Strength (psi)	12,200	18,900	11,000
D638	Tensile Modulus (psi)	385,000	825,000	340,000
D638	Tensile Elongation at Break (%)	6.5	1.9	30
D790	Flexural Strength (psi)	16,100	23,500	15,500
D790	Flexural Modulus (psi)	420,000	950,000	345,000
D695	Compressive Strength (psi)	14,500	25,600	13,400
D695	Compressive Modulus (psi)	388,000	1,119,000	280,000
D785	Hardness, Rockwell	M88 / R127	M80 / R124	M80 / R120
D256	IZOD Impact Notched (ft-lb/in)	1.6	1.4	2.5
THERMAL				
D696	Coefficient of Linear Thermal Expansion (x 10 ⁻⁵ in./in./°F)	2.70	1.70	3.10
D648	Heat Deflection Temp (°F / °C) at 264 psi	400 / 204	420 / 215	405 / 207
D3418	Melting Point Temp (°F / °C)	- / -	- / -	428 / 220
-	Max Operating Temp (°F / °C)	320 / 160	320 / 160	180 / 82
C177	Thermal Conductivity (BTU-in/ft ² -hr-°F) (x 10 ⁻⁴ cal/cm-sec-°C)	1.13 3.89	1.35 4.65	2.4 8.3
UL94	Flammability Rating	V0	V0	V-O
ELECTRICAL				
D149	Dielectric Strength (V/mil) short time, 1/8" thick	380	440	360
D150	Dielectric Constant at 1 MHz	3.5	4.1	3.44
D150	Dissipation Factor at 1 MHz	0.0022	0.0018	0.0017
D257	Volume Resistivity (ohm-cm) at 50% RH	1.7 x 10 ¹⁵	> 10 ¹⁶	> 10 ¹⁵

Data provided is for reference purposes only. Additional testing may be required for design specifications or quality control. All values at 73°F unless otherwise stated.

